

Enhanced DETR-Based Framework for Automated Wood Chip Size Distribution Estimation in High Volume Biomass Manufacturing

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Background

Wood chip geometry plays a critical role in biomass manufacturing.

Efficient Combustion

Uniform chips burn better with less waste.

Product Quality

Uniform size leads to stronger, smoother paper.

Improved Pulping

Consistent size = better chemical penetration.

Smooth Handling

Reduces clogs and flow issues in processing lines.

Current Challenges in Wood Chip Size Estimation:

- Oscillating screen method $\longrightarrow$ Slow, Labor-dependent, energy-intensive
- Scalability issues $\longrightarrow$ Thousands of chips per batch
- Manual annotation $\longrightarrow$ Lack of existing large datasets

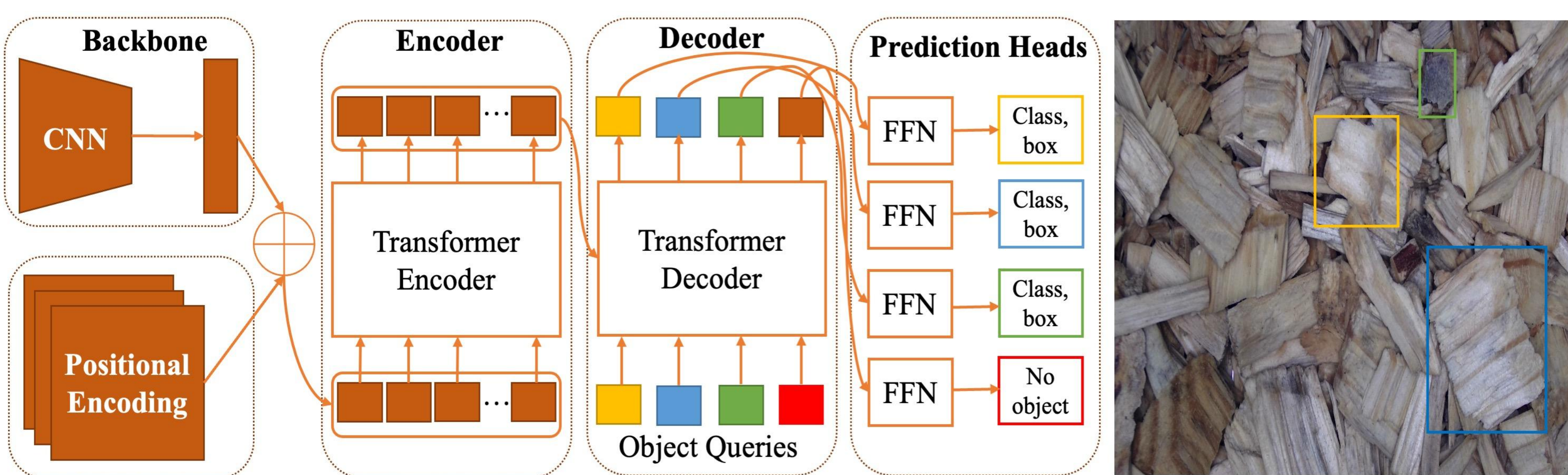
“Can an advanced machine learning (ML) model provide reliable wood chip detection and size estimation for industrial biomass processing?”

Objectives

- Develop an ML model for detecting small, overlapping, and irregular wood chips.
- Create a diverse, annotated dataset with over 7,100 chips from multiple mills.
- Estimate chip size distributions and validate with KL divergence.
- Apply post-processing filters for refined detection.

Methodology

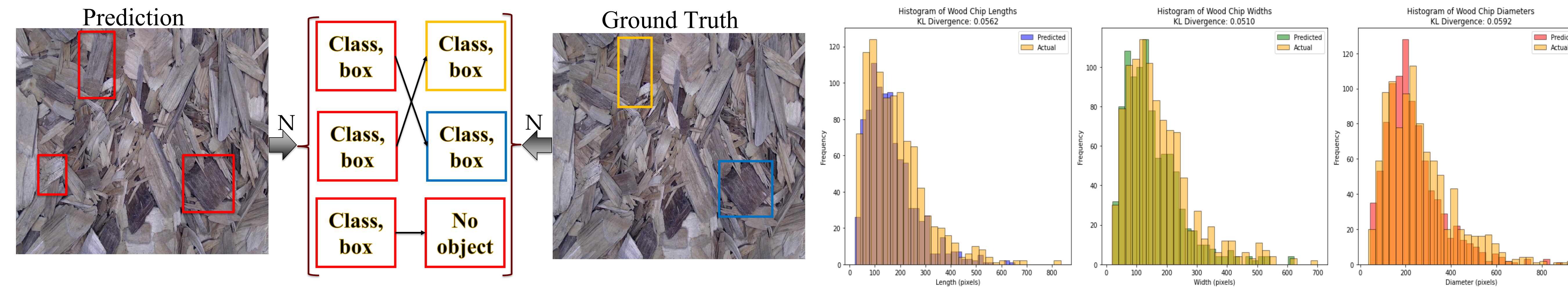
Enhanced DETection TRansformer (DETR)



Architecture of the proposed enhanced DETR-based model.

Bipartite matching cost

$$\hat{\sigma} = \arg \min_{\sigma \in \mathbb{S}_N} \sum_{i=1}^N L_{match}(y_i, \hat{y}_{\sigma(i)})$$



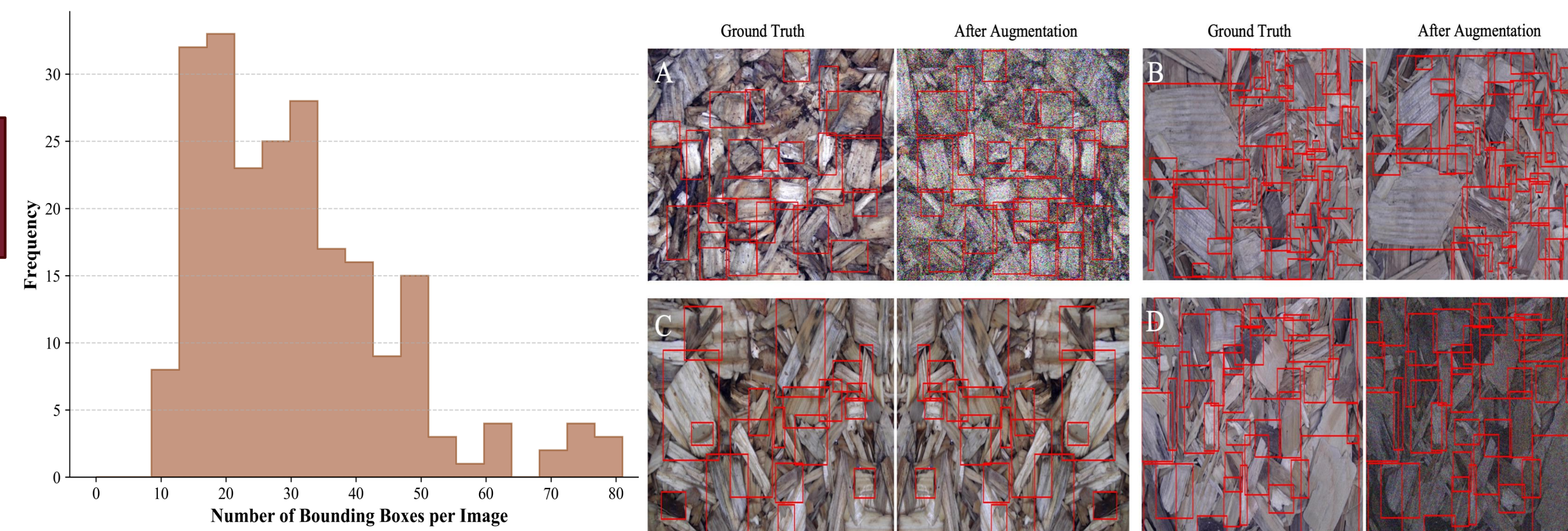
Histogram of Predicted vs. Ground Truth Chip Size Distributions

Hungarian Loss:

$$L_{Hungarian}(y, \hat{y}) = \sum_{i=1}^N [-\log \hat{p}_{\hat{\sigma}(i)}(c_i) + 1_{\{c_i \neq \emptyset\}} L_{box}(b_i, \hat{b}_{\hat{\sigma}(i)})]$$

Dataset description:

The dataset: 223 images with over 7,100 annotated wood chips.



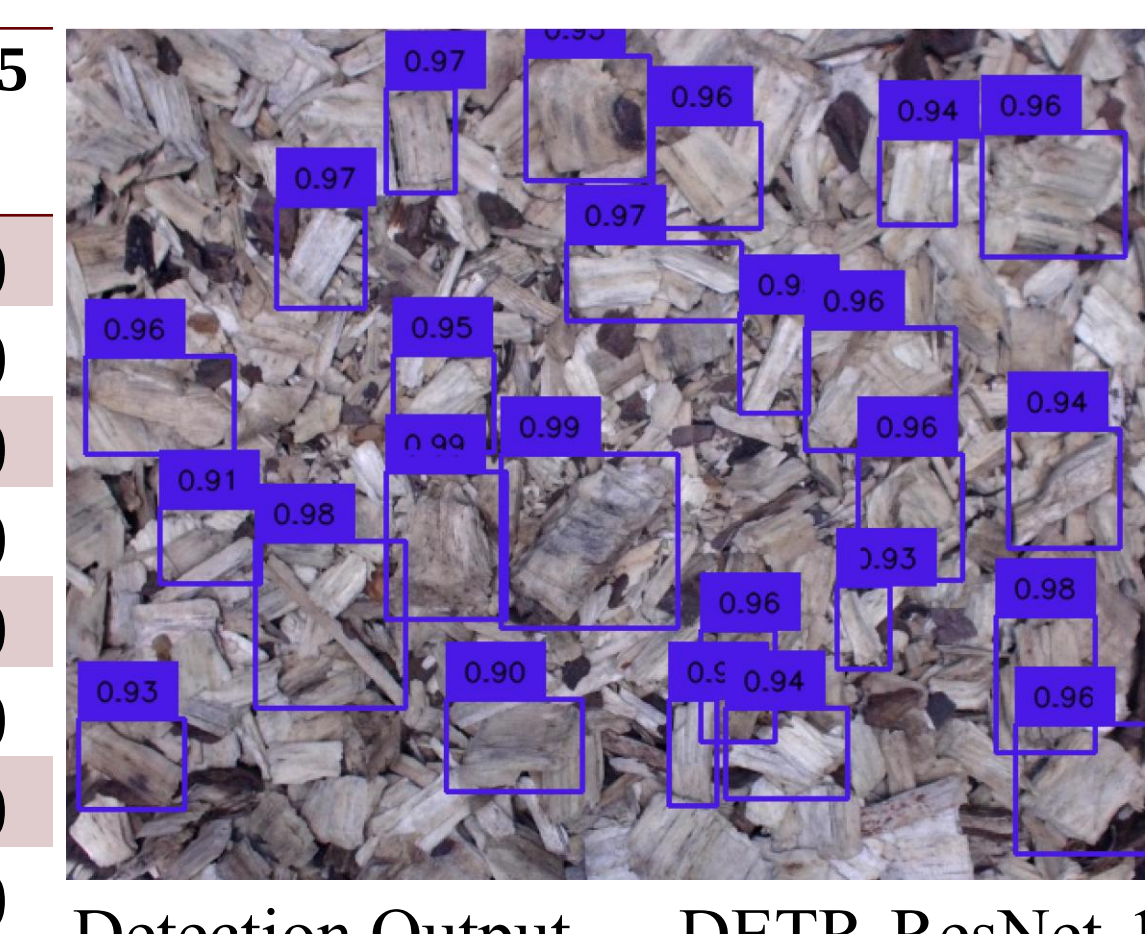
Experimental Results

Performance Metrics (5-fold CV)

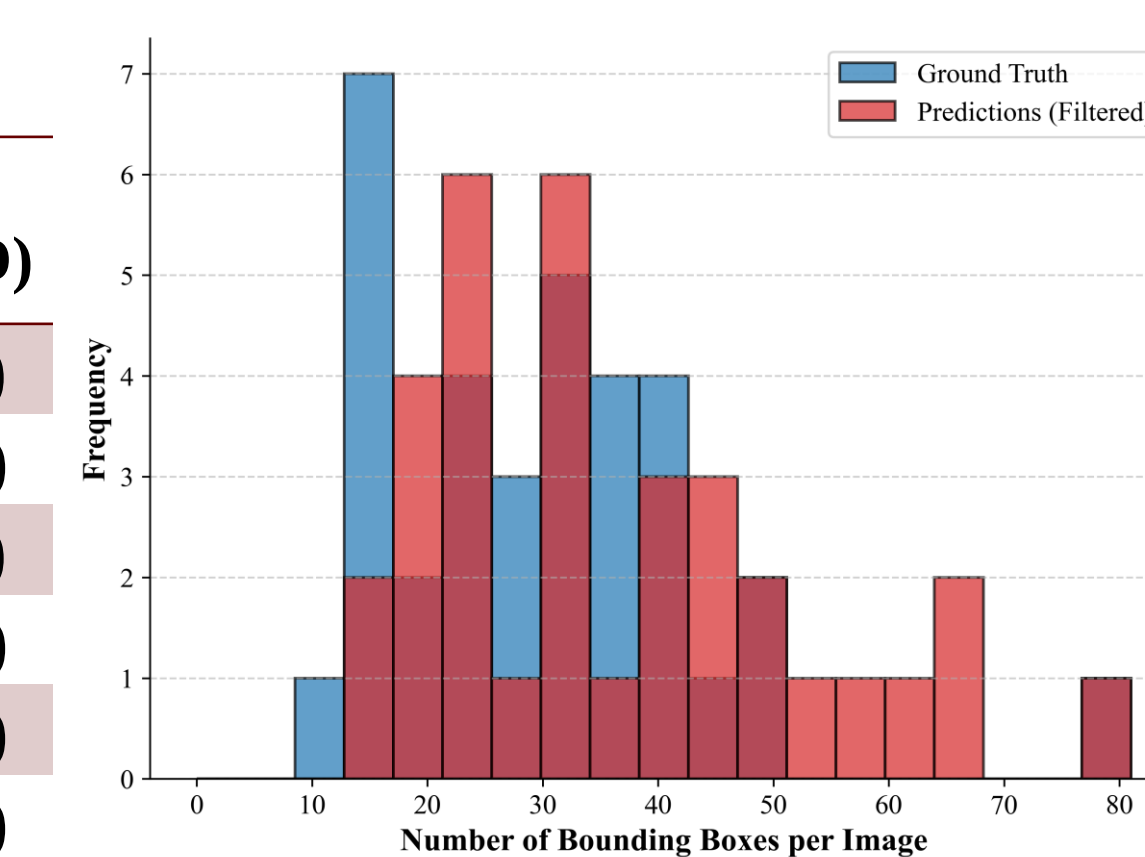
Model Architecture	mAP50 (SD%)	mAR50 (SD%)	mAP50-95 (SD%)	mAR50-95 (SD%)
Faster RCNN R50-FPN	57(1.02)	59(1.85)	39(0.98)	51(2.04)
Faster RCNN R101-FPN	57(2.58)	59(2.64)	40(1.50)	49(2.04)
Faster RCNN R101-DC5	60(1.62)	62(2.04)	42(1.14)	54(2.32)
Retinanet R50-FPN	63(1.41)	68(2.48)	41(0.75)	62(1.10)
Retinanet R101-FPN	65(1.62)	72(2.33)	43(1.50)	62(0.80)
DETR-ResNet-50	76(0.89)	85(0.49)	51(0.75)	57(1.10)
DETR-DC5-ResNet-101	80(1.89)	85(2.16)	54(3.40)	60(4.32)
DETR-ResNet-101	85(2.28)	91(1.72)	61(3.77)	68(3.25)

Kullback-Leibler Divergence

Model Architecture	Length KL divergence (SD)	Width KL divergence (SD)	Diameter KL divergence (SD)
Faster RCNN R50-FPN	0.066(0.006)	0.058(0.006)	0.075(0.011)
Faster RCNN R101-FPN	0.082(0.008)	0.079(0.008)	0.073(0.009)
Faster RCNN R101-DC5	0.080(0.010)	0.069(0.007)	0.077(0.011)
Retinanet R50-FPN	0.068(0.011)	0.067(0.001)	0.085(0.021)
Retinanet R101-FPN	0.077(0.004)	0.070(0.007)	0.093(0.012)
DETR-ResNet-50	0.070(0.012)	0.078(0.014)	0.080(0.012)
DETR-DC5-ResNet-101	0.065(0.004)	0.100(0.016)	0.083(0.007)
DETR-ResNet-101	0.064(0.007)	0.057(0.015)	0.073(0.012)



Detection Output — DETR-ResNet-101



Bounding Box Distribution in Test Dataset

Conclusions

- Developed an enhanced DETR-based pipeline for real-time wood chip size estimation in high-volume biomass manufacturing.
- Accurately detects small, overlapping, and irregular chips by integrating self-attention, bounding box filtering, IoU-based validation, and confidence refinement.
- Trained on over 7,100 annotated chips, the model achieved mAP50 = 85%, mAR50 = 91%, mAP50–95 = 61%, and mAR50–95 = 68%.
- Future work** - improve efficiency by refining the backbone and speeding up inference for real-time biomass processing.

Selected References

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